

Cultural overlap does not reliably reflect connectivity: The differential spread of adaptive and neutral traits

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Abstract

Inferring the relations between cultures from material artifacts in the archaeological record is one of the major challenges in prehistoric archaeology. Understanding the dynamics of spread of cultural traits, such as tool types and production techniques, is crucial for such inference. Many studies highlight inter-population connectivity as a major determinant of such spread. The effect of traits' characteristics, on the other hand, on their probability to spread, is rarely considered. Here, to gain insight into the spread of cultural traits, we use analogies with theory in population genetics, specifically regarding the spread of genetic traits as a function of their fitness consequences. The theory, rather intuitively, predicts different probabilities of spread for neutral and adaptive traits. Useful predictions can be derived from this simple differentiation. Using a dynamic computational model, we explore the differential dynamics of traits and show that: (1) At the same levels of connectivity, selectively adaptive traits are more likely to spread widely, while neutral traits may remain localized (2) Thus, ignoring the differential spread probabilities of different traits can lead to a misinterpretation of connectivity; (3) Similarly to reconstructing genetic phylogenies, selectively neutral traits may be useful in reconstructing the relationships between closely related populations, while adaptive traits could be suitable for identifying more distant relationships.

Introduction

The study of interactions between populations, present and past, is a fundamental aspect of both anthropology and archaeology. When exploring a new archaeological site, researchers often ask how its material assemblage relates to those at other sites: Was it connected at all to other sites, and if so, to what extent? What was the nature of the connection? Do they share an origin, or was there later contact? (See Foley and Lahr, 2003; Malinsky-Buller & Hovers, 2019; Trigger, 2006; Goren-Inbar et al., 2000; Hovers & Belfer-Cohen, 2013). These questions are vital in reconstructing populations' history but also in understanding how their culture evolved (Creanza, Kolodny, and Feldman, 2017; Greenbaum et al., 2019; Ben-Oren et al., 2023). This is because inter-population contact can not only allow cultural (and genetic) exchange but also facilitate the emergence of new ideas or cultural combinations.

A common method for estimating the connectivity between populations is to examine the cultural similarity between them. For example, when comparing two cultural assemblages, the fraction of cultural overlap (i.e., shared cultural traits) can be used as a proxy for inter-population connectivity. A recent example of such an approach is Jiménez-Puerto, 2024, who, using Social Network Analysis, studied the distribution of arrowhead types in the east of the Iberian Peninsula between the Late Neolithic and Early Bronze Age (5300-3800 years BP), identified periods of cultural homogenization and cultural differentiation across sites, and interpreted them, correspondingly, as periods of high and low connectivity.

We suggest that the use of cultural overlap as a proxy for inter-population contact may lead to misinterpretations of past connectivity. This is because the probability of a cultural trait being shared between populations is not only determined by the strength of contact between them, but also by the trait's own characteristics – a commonly overlooked factor. In other words, under the same connectivity conditions, some traits may readily spread, and become shared across multiple populations, while others may barely spread beyond their population of origin, portraying what would seem to be a very different map of population connectivity. The effect of the characteristics of cultural traits on their sharing across populations has recently been demonstrated in non-human primates: Gunasekaram et al. 2024, reported distinct cultural overlap patterns in wild chimpanzee populations when considering simple versus complex traits.

We apply population genetics theory to study the spread of cultural traits. The analogy between genes and cultural traits has been used in various studies of cultural evolution (Cavalli-Sforza & Feldman, 1978; Boyd & Richerson, 1985; Fogarty et al., 2024). From these studies, we know that the spread of genetic alleles (analogous to cultural variants) depends on their adaptive value. Based on Kimura (1962), we estimate that the probability of a newly introduced, selectively neutral allele taking over a population is proportional to its relative frequency ($1/N$ for a single allele in a haploid population of size N). In contrast, an adaptive allele's probability of fixation is proportional to its selection coefficient ($\sim 2s$ for a single allele with selective advantage s). For example, a neutral allele has a 0.01% chance to dominate a population of 10,000 individuals, while an allele with $s = 0.05$ has a 10% chance (1000 times higher).

Drawing from phenomena described in population genetics, we can predict that under low or intermediate connectivity rates, selectively neutral cultural traits will have very low probabilities of spreading between populations, while selectively adaptive traits will have much higher probabilities. This raises an important question of what should be considered an adaptive trait in cultural evolution. While in genetics, the adaptive value of a trait is quantified by its contribution to the carrier's survival and reproductive success, the picture in culture is more complex. This is because the fitness of a cultural trait can be affected by many other factors, such as its attraction (Henrich & Boyd, 2002; Claidière & Sperber, 2007), "stickiness"/"retention potential" (Strimling, Enquist, & Eriksson, 2009), or how easy it is to learn (as mentioned above in Gunasekaram et al., 2024; Truskanov & Lotem, 2017). While there are nuanced differences in how these factors influence cultural transmission, and despite key distinctions between genetic and cultural evolution (see Discussion), we suggest that the concept of cultural fitness remains a practical unifying framework for capturing the differential success of cultural traits in a population (Ramsey & De Block, 2017; Henrich, Boyd, and Richerson, 2008; Mesoudi & Thornton, 2018).

In this paper, we use a spatially explicit computational model to simulate cultural transmission among partially connected populations. We begin by exploring the differential spread patterns of adaptive and neutral cultural traits. Then, we compare the relationships between geographical distance and resulting cultural overlap in adaptive and

neutral trait categories. Finally, we use our model to reconstruct the connectivity between simulated populations based on adaptive and neutral trait sharing patterns, demonstrating its sensitivity in identifying different connectivity rates.

The model

We propose a model for the spread of cultural traits (hereafter referred to as “tools” or “variants” for simplicity), derived from models of population genetics. In the model, we monitor different “adaptive” and “neutral” tool categories (equivalent to genetic loci under selection and neutral loci), for which each population is characterized by a dominant tool variant (equivalent to a genetic allele or variant). variants can change over time only through a single process – replacement by other variants, introduced either by either migration or local invention. The probability of fixation of newly introduced **neutral** variants in a population (=replacement of the existing variant) is a product of random drift alone. Thus, if such a tool is carried by a single immigrant or inventor in a population of size N , its fixation probability would be $1/N$ (as each individual has an equal chance of being the one whose variant eventually spreads stochastically to the whole population). For variants in **adaptive** categories, on the other hand, the probability of fixation is dependent on their selective advantage **in relation to the preexisting dominant tool**. Hence, if the newly introduced tool, carried by a single individual, has a selective advantage of size s over the prevalent tool, its fixation probability would be $\sim 2s$, regardless of population size (see Introduction for elaboration). Importantly, when two adaptive variants have equal selection coefficients, their dynamics will mirror those of two neutral variants.

To explore the dynamics generated by these assumptions, we propose a spatially explicit simulation to explore how adaptive value affects the spread of cultural traits between culturally interconnected populations. The simulations consist of a 10x10 grid of populations, each containing 1000 individuals. The cultural repertoire of each population is monitored throughout a simulation run lasting 1000 time units, in which tool variants in each population can be invented, transmitted, and replaced. The simulation has two versions, one in which variants of a single tool category are monitored (i.e., for example, the type of knife prevalent in each population, where each population has a single dominant

type at each time unit), and another, where entire cultural repertoires, of 20 tool categories, are simulated simultaneously. Under each simulation version, we simulate cultural change in selectively neutral and adaptive tool categories.

At the beginning of each simulation, a unique variant is assigned to each population within each simulated tool category. For non-neutral tool categories, each variant is assigned a selection coefficient drawn from an exponential distribution with $\beta = 0.1$, similar to Kolodny et al. (2015). In each time unit, populations' cultural repertoires can change due to inventions and cultural transmission from neighbors as an outcome of migration.

Variant substitutions due to local invention occur at a probability of 10^{-4} per population. In adaptive tool categories, new tools are assumed to be “developments” of existing variants, although counted as distinct variants, and are therefore assigned a selective advantage drawn from an exponential distribution with $\beta = 0.1$, which is added to the selection coefficient of the existing tool. Importantly, while neutral traits may be expected to be replaced more frequently within populations than adaptive traits, we conservatively assumed they change at the same pace (see the Discussion for elaboration). Importantly, however, assuming a higher rate of local substitutions for neutral traits would only strengthen the results.

At the population level, new tools can also be introduced by migrants from neighboring populations. The number of migrants leaving each population is drawn from a Poisson distribution with $\lambda = 1$ (similarly to Ben-Oren et al. 2023), and their destination is randomly chosen among the immediate neighbors.

Whenever a new variant is introduced to a population, either by invention or migration, we assume its probability of fixation is dependent on its selective advantage (again, $1/N$ for neutral tools and $2s$ for adaptive tools, where s is the selective advantage of the new variant over the existing one).

Notably, it is admittedly nearly impossible to estimate exact parameter values (e.g., population sizes, migration rates between sites, probabilities of invention, etc.) from archeological data. However, since, as explained in the Introduction, the differences in the

probabilities of spread between adaptive and neutral cultural traits are qualitative, and therefore independent of specific parameter choices. Because this work aims to illuminate the general implications of these differences, rather than to explore a specific case, we can thus choose parameter values arbitrarily.

We begin by simulating the dynamics for a single tool category (which is either adaptive or neutral). This allows us to easily visualize the spread of specific variants and the overall metapopulation patterns. Then, we simulate multiple categories of the same kind (neutral/adaptive) simultaneously. This helps us investigate the relationship between inter-population connectivity and the similarity of cultural repertoires in neutral and adaptive categories. Because in our default model, connectivity between all neighboring populations is identical, i.e., populations are isolated by distance, we can use physical distance (more precisely, Chebyshev distance, i.e., the minimal number of horizontal, vertical or diagonal steps between population coordinates) as a proxy for network distance and ask how well the geographical map is reflected in the cultural repertoires of the different populations. We begin by comparing the cultural overlap in adaptive and neutral tools as a function of geographical distance, and then use a UMAP analysis (a dimensionality reduction technique similar to PCA) to attempt to reconstruct the metapopulation map from the neutral and adaptive cultural repertoires for different rates of migration (i.e., interpopulation connectivity). Notably, while we chose, as a proof of concept, to use UMAP rather than other similar techniques because it is particularly suitable for preserving topological structures and pairwise distances (McInnes, Healy & Melville, 2018), PCA yielded qualitatively similar results.

Results

We start by monitoring the dynamics for a single tool category across the metapopulations. Assuming maximal cultural diversity at the beginning of each simulation (i.e., that each population has a unique variant), we find that for neutral tool categories, the spread of specific tools is highly restricted. That is, under the given conditions, even the most common tools did not exceed 10% of the metapopulation by the end of the simulation, creating small patches of shared variants (Figure 1).

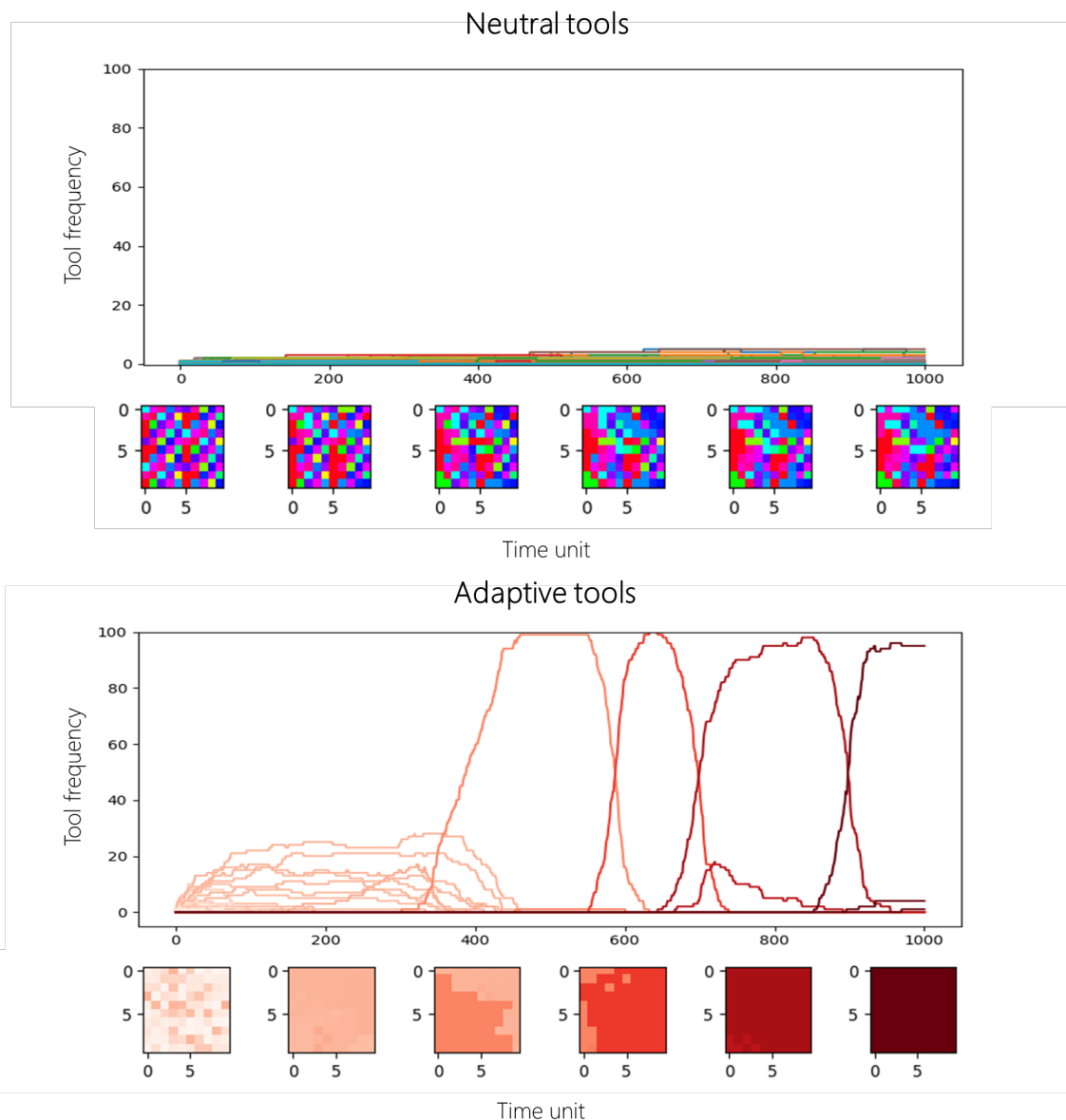


Figure 1 At the same rates of interpopulation connectivity, adaptive tools can spread efficiently, while neutral tools remain localized. In each graph, each line signifies the percentage of populations sharing a specific variant throughout the simulation. At the bottom of each graph, maps of variants found in each population were plotted at different time points, with variant colors corresponding to those in the graph. For neutral tools, each variant was assigned a color randomly. For adaptive tools, each variant was colored according to its selection coefficient, with higher selection coefficients represented in darker shades. Under the conditions explored, neutral tools spread very minimally, with no variant becoming particularly common, while for adaptive tools, variants with high selection coefficient can spread very rapidly, and even take over the entire metapopulation.

For an adaptive tool category, the same conditions yielded qualitatively different results (Figure 1). In this scenario, the dynamic can be broadly divided into two phases. In the first

197 phase (observed in the figure roughly between time units 0 and 300), variants with low
198 selection coefficients are replaced by variants with higher selection coefficients. The
199 surviving variants, all sharing a similar, relatively high selection coefficient (hence similar
200 shades on the graph), are then subjected to neutral drift and, therefore, coexist for an
201 extended period of time. Around time unit 300, a new variant with a significantly higher
202 selection coefficient is invented in one of the populations and starts spreading rapidly.
203 Although nothing changed in the metapopulation's interconnectivity, this variant fixates in
204 the metapopulation, replacing all other variants. Later, variants with increasingly higher
205 selection coefficients emerge and repeatedly replace one another in a similar fashion.

206 Next, we simulate the transmission dynamics for entire cultural repertoires consisting of
207 20 tool categories. For these simulations, we examined how physical distance (i.e., the
208 number of steps between populations) is reflected in the similarity in cultural repertoires
209 for neutral and adaptive toolkits (see Figure 2). We calculated the fraction of cultural
210 overlap between all couples of populations with geographical distances of 1 (immediate
211 neighbors), 2, 4, and 8. We find, as expected, that regardless of tools' adaptiveness,
212 geographical proximity contributed to cultural similarity. However, we also find that for
213 neutral tool categories, even immediate neighbors shared fewer tools (17% on average)
214 than populations with distance 8 did for adaptive categories (63% on average). Moreover,
215 we find that the decay in cultural similarity as a function of distance is much more rapid
216 for neutral toolkits than it is for adaptive toolkits. For neutral toolkits, the cultural overlap
217 between neighbors with distance 2 (3%) was significantly lower than for neighbors with
218 distance 1 (17%), and more distant relationships left no traces. For adaptive toolkits, the
219 cultural overlap remained similar for distance 1 (87%) and 2 (83% overlap), with most
220 tools being shared even between neighbors of distance 8.

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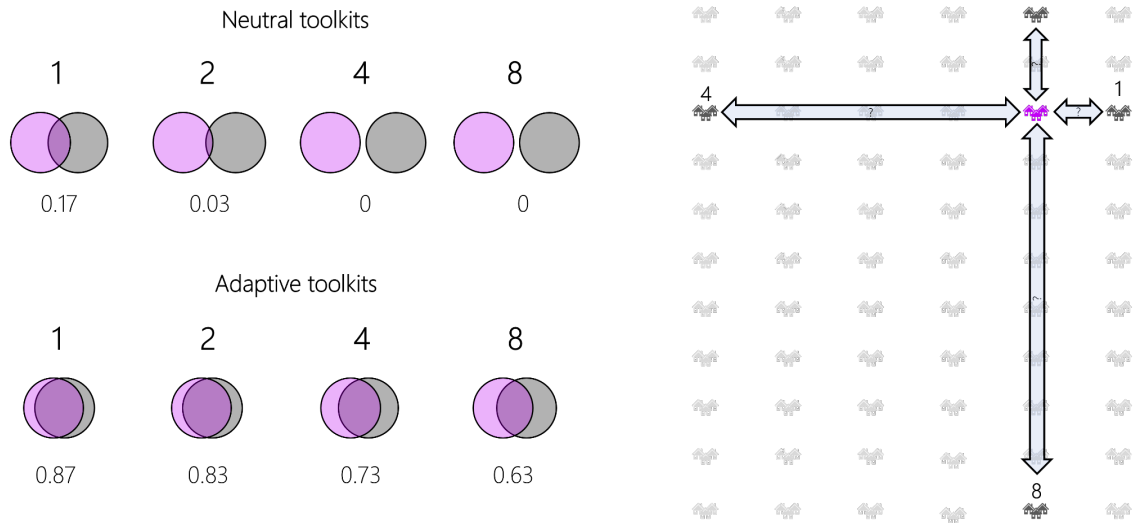


Figure 2 Neutral tools are suitable for distinguishing between high levels of connectivity, while adaptive tools are useful for detecting distant relationships.

Finally, to demonstrate the different resolutions of connectivity signals conveyed by neutral and adaptive tools, we try to reconstruct the metapopulation map (reflecting the connectivity map in our simulations) using a UMAP analysis based on either adaptive or neutral tool repertoires (Figure 3) for different rates of interpopulation connectivity (migration probabilities). For higher rates of migration ($\lambda = 1$, i.e., one migrant per population per time unit, the default value used in all simulations), neutral toolkits yielded more accurate reconstructions, while for lower rates of migration ($\lambda = 0.01$), adaptive tools yielded more accurate reconstructions. This is because when connectivity is relatively high, adaptive tools spread very efficiently and tend to be shared throughout the metapopulation, regardless of distance (see also previous results), while neutral tools tend to be shared only between close neighbors. When connectivity is very low, adaptive tools spread more slowly, and neutral tools cannot spread at all.

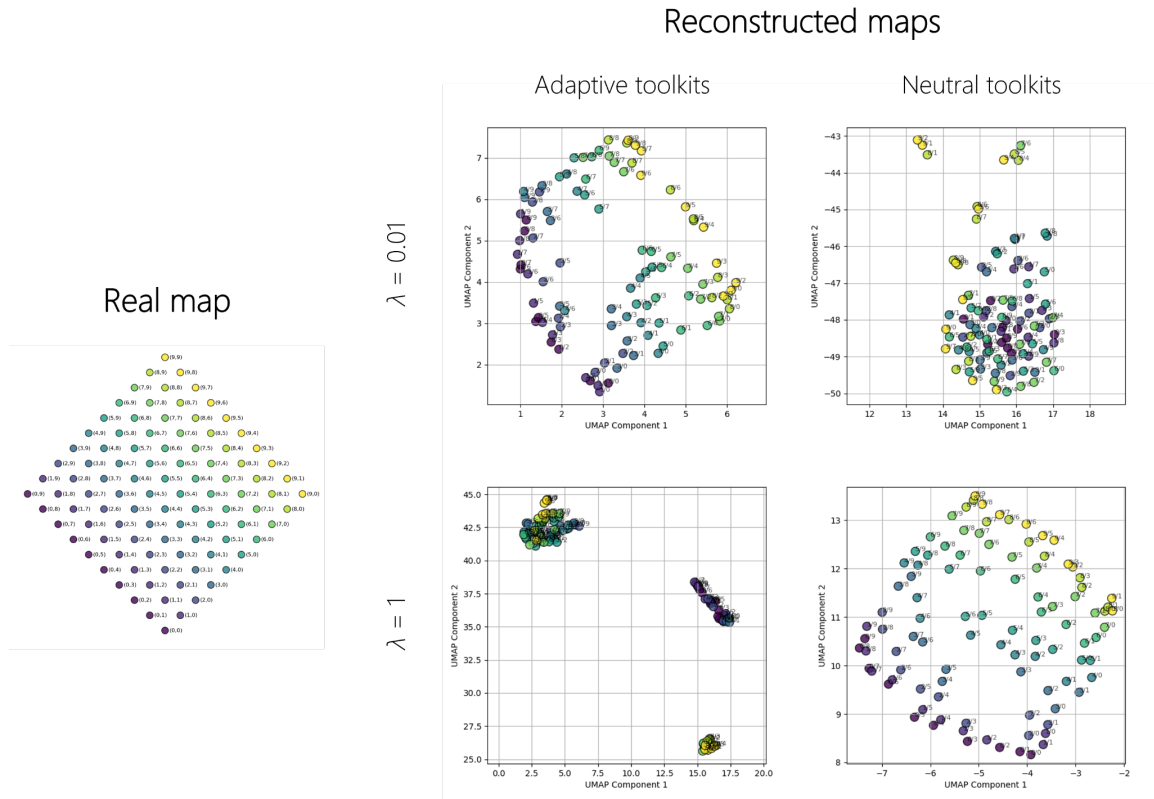


Figure 3 Neutral toolkits are more informative for reconstructing relationships between close neighbors, while adaptive toolkits are more suitable for reconstructing distant relationships. Using UMAP analysis, we aimed to reconstruct the metapopulation map (in our case, equivalent to the connectivity map), presented in panel A (rotated for easier comparison to the UMAP plots) using adaptive and neutral tool repertoires, under different rates of connectivity.

Discussion

Our findings highlight the fundamental differences between the transmission dynamics of different cultural traits within the same connectivity networks. They demonstrate that the probability of spread of a cultural trait is highly dependent on its characteristics – in our model captured by the distinction between adaptive and selectively-neutral cultural traits. We suggest that adaptive cultural traits are much more likely to spread between populations, even when connectivity is limited. In contrast, unless connectivity is extremely high, neutral cultural traits tend to remain localized, leading to higher levels of cultural diversity across populations. This means that the same environment may be “experienced” as highly connected by some cultural traits and highly fragmented by others. This has important implications for interpreting the archaeological record, as it suggests that cultural similarities observed between populations in different cultural categories may

be indicative of selective pressures rather than a direct reflection of inter-population connectivity. Because the archaeological record is often limited, we aim to highlight the qualitative insights our model can offer even with incomplete information, while also illustrating how more detailed data can enable richer, quantitative interpretations. Returning to the example of Jiménez-Puerto 2024, mentioned in the Introduction, a temporal change in cultural similarity for a single tool category (in their case, arrowhead type) may not necessarily result from a change in connectivity, but rather from a shift in selection dynamics – e.g., from periods of neutral dynamics between variants with similar adaptive value to occasional emergences of highly adaptive variants that sweep across the population.

Our findings suggest that different categories of cultural traits can provide varying resolutions for reconstructing past connectivity patterns. Neutral cultural traits, due to their limited spread, are highly sensitive to fine-scale connectivity differences and thus offer the most precise reconstruction of close relationships. In contrast, highly adaptive traits, which can travel long distances even under low connectivity, are better suited for identifying more networks that are more spatially extensive. This suggests that to reconstruct nuanced variations in connectivity rates between neighboring populations, archaeologists should prioritize neutral cultural traits, while to identify looser, long-range connections, adaptive traits provide a more informative signal. This distinction is used in reconstructing genetic phylogenies, as genetic loci under selection tend to be more conserved and similar even across distant species, while selectively neutral loci tend to be more variable, making them informative for reconstructing closer relationships.

By examining the spread patterns of adaptive and neutral traits, we can, given the relevant data, infer both past connectivity and the adaptive value of certain cultural elements. If a cultural trait is widespread despite evidence of low connectivity (e.g., because of natural or cultural barriers), it is likely adaptive. Conversely, if a cultural trait remains highly localized despite high connectivity, it may be neutral (or locally adaptive; see below). This reciprocal relationship suggests that archaeological interpretations should, whenever possible, consider both cultural transmission dynamics and functional advantages when analyzing trait distributions. Alternatively, if the cultural record is sufficiently rich and

includes traits that undergo varying selection dynamics, it may also be possible to reconstruct connectivity maps using all traits collectively.

While we designed our model to be as simple and tractable as possible and presented only a single, simplistic scenario (with all parameters being uniform throughout space and time, tools being independent of one another, etc.), we believe that the framework presented can provide insight into a wide range of empirical and theoretical cases.

In our model, tools were divided, for simplicity, into two categories – adaptive and neutral. However, other distinctions between tools can be useful. These, for example, include adaptive tools with different selection coefficients or different adaptive features (e.g., attractiveness vs. contribution to individual/group fitness), traits that favor diversification like style (Garvey, 2018; Youngblood, Miton, & Morin, 2023), and linked traits with different selection coefficients (Herzlinger & Goren-Inbar, 2020; Crema et al, 2016; Yeh et al., 2021; Carrignon et al., 2024; Henrich, Boyd, & Richerson, 2008).

Our simulations assume, for simplicity, that the rate of trait substitution within populations (i.e., emergence and replacement of new variants) is the same for adaptive and neutral traits. As discussed in The Model section, this is not necessarily the case. Empirical evidence (Rogers & Ehrlich, 2007) suggests symbolic traits may change faster than functional traits, aligning with the expectation that neutral traits will emerge more frequently than adaptive traits (similar to neutral mutations occurring more frequently than adaptive mutations). If this is the case, it would enhance the difference in cultural overlap between neutral and adaptive traits, making neutral traits even more localized. Although the claim that neutral cultural traits change more rapidly is supported by some empirical evidence, as well as by the theoretical expectation for a higher frequency of emergence of neutral traits, we chose to assume that they change at the same pace conservatively. That is because the likelihood of establishment per trait is higher for adaptive traits, which may counteract their lower frequency. Other effects that may contribute to higher cultural similarity in adaptive traits and are not explored here include convergence and spontaneous cultural loss. Adaptive traits tend to converge culturally because similar problems in different societies often lead to similar, selectively favored solutions, whereas neutral traits

drift without such direction, leading to higher similarity in adaptive traits. Similarly, while our model simulates cultural loss solely as a result of replacement by other variants, in reality, cultural loss may occur independently of the introduction of new cultural elements, and such “spontaneous” loss may be more likely to happen in neutral traits. Hence, neutral traits may be less persistent and therefore, even less likely to be shared across populations.

As mentioned above, while here we explore only a scenario where all parameters are uniform across space and time, many other, more complex settings are worthy of exploration. For example, a spatial difference in the adaptive value of a trait may create adaptive “valleys” that prevent the spread of cultural traits that are functional only in certain locations. Similarly, adaptive values can change, not only through space, but also over time. An interesting case study where we suggest this could be the case is the emergence of symbolism (Bar-Yosef, 1997; Hovers & Belfer-Cohen, 2022; Stibbard-Hawkes, 2025), where it has been recently suggested (Ben-Oren et al., 2025) that its sudden spread, after a long period of anecdotal appearances, could be a change in selection forces.

Notably, while in our simulations every population is initially defined as culturally distinct, and cultural similarity is only a product of diffusion between populations, in reality, cultural similarity can also arise from shared ancestry (Mace & Holden, 2005; Goren-Inbar et al., 2000). Our model predicts that such cases will be identified by exceptional overlaps in neutral cultural repertoires that are not expected otherwise.

Our findings highlight low and intermediate connectivity as unique situations where neutral variation can be preserved while adaptive traits continue to spread. This may allow for the spread of inventions of high utility while still maintaining much of the society’s cultural diversity and “evolutionary potential” (See Supplementary Materials S1; Wagner 2023). While, as discussed above, cultural traits that are selectively adaptive will not necessarily be adaptive to their carriers, traits that are highly beneficial are likely also to be culturally selected. This finding is consistent with other studies that identify low or intermediate levels of connectivity as contributors to cultural accumulation for different reasons (Derex & Boyd, 2016; Derex, Perreault, & Boyd, 2018) and the resilience of natural populations (Morjan & Rieseberg, 2004).

Interestingly, the difference between the spread patterns of adaptive and neutral traits is expected to be affected by (effective) population size. Theory suggests that the probabilities of fixation of adaptive alleles depend on population size, particularly in small populations (Kimura, 1962). Therefore, given sufficient data resolution, comparing the spread patterns of adaptive and neutral cultural traits can help us reconstruct prehistoric demography (see Supplementary Materials S2). However, a key distinction in making these kinds of evaluations is that they reflect effective population sizes rather than census sizes. In other words, if a large population includes only a small number of decision-makers who decide which cultural practices to adopt, the effective cultural population size (in terms of stochasticity in cultural change) will be small. This can potentially help us understand within-group cultural dynamics.

In conclusion, this study contributes to an expanding line of research employing population genetics as a theoretical lens for understanding cultural change, underscoring the explanatory potential of this framework. While we caution against using cultural overlap between populations as a direct proxy for connectivity, we suggest that accounting for the differential spread dynamics of different cultural traits can facilitate an informed interpretation of past inter-population contact and cultural dynamics.

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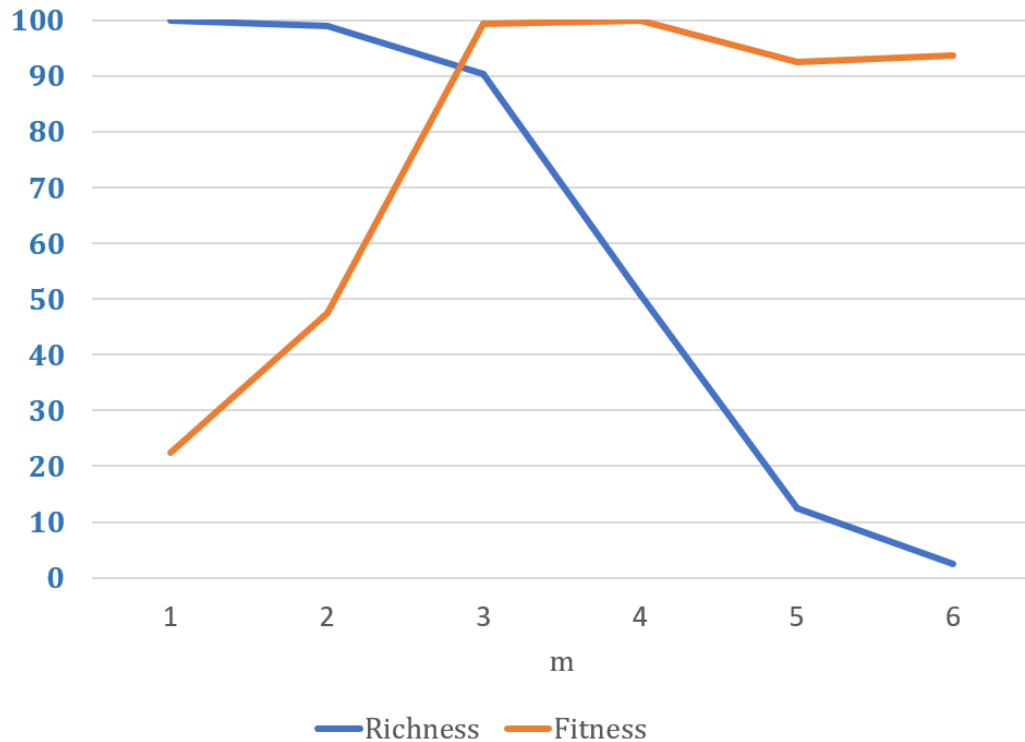
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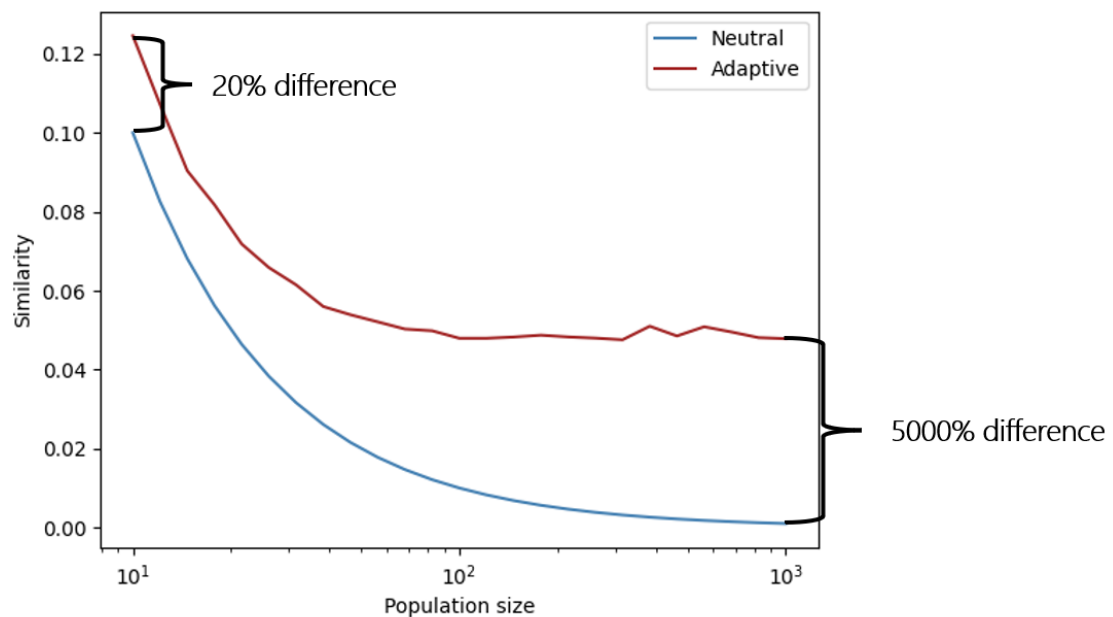
Supplementary Materials

S1 – Intermediate connectivity allows for the spread of cultural adaptations while conserving neutral cultural diversity



Metapopulation neutral cultural richness (=total number of neutral tools) and average cumulative fitness (=sum of the selection coefficients of all adaptive tools in a population) as a function of the rate of migration between populations. The y-axis is normalized, so 100 signifies the maximal cultural richness and maximal cumulative fitness achieved through all simulations. We show that This has important implications for interpreting the archaeological record low migration rates allow for the spread of adaptations while maintaining most of the metapopulation's neutral cultural diversity.

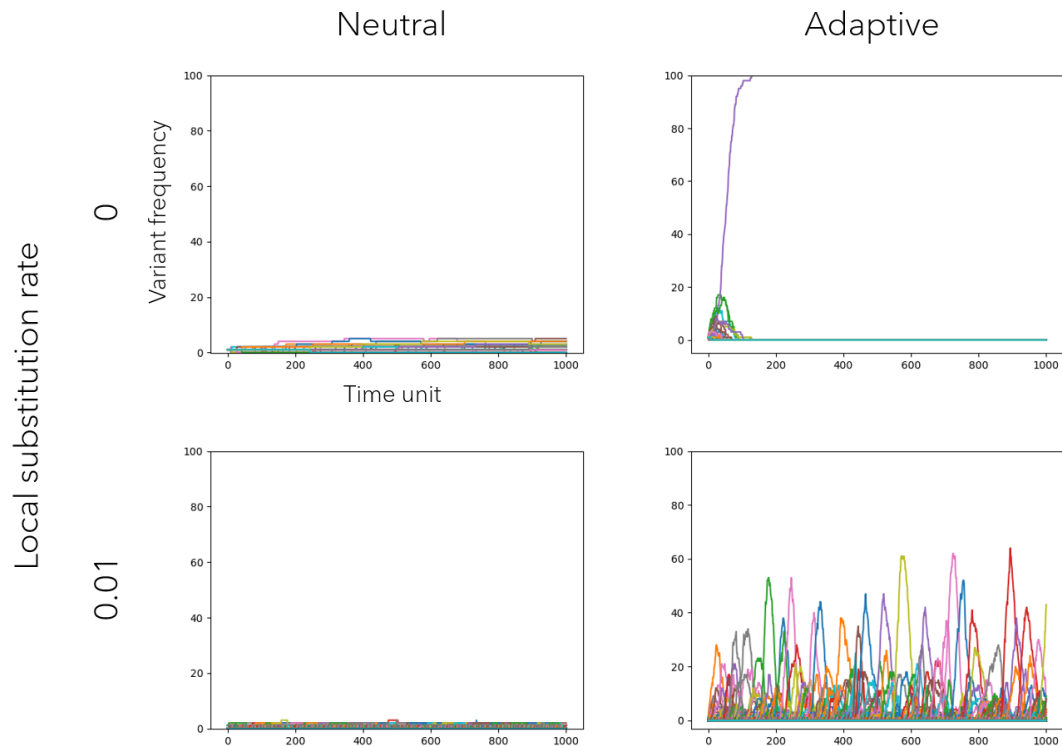
S2 – The difference between the spread patterns of neutral and adaptive tools reflects effective demography



The difference in cultural overlap between neighboring populations for adaptive and neutral tool categories as a function of population size. When populations are (effectively) small ($n \cdot s \ll 1$), selection coefficient has little effect on fixation probabilities, thus leading to similar rates of cultural overlap across categories. while when they are sufficiently large ($n \cdot s \gg 1$), the difference between the fixation probabilities of neutral and adaptive tools can be of several orders of magnitude, leading to big differences in cultural similarity between adaptive and neutral tool categories.

S3 – Differential spread patterns of neutral and adaptive tools are robust to changes in the rate of local substitutions

Tool Category



We examine how changing the rate of local substitutions (emergence and replacement of variants within populations) affects the spread patterns of neutral and adaptive tools. Under the conditions explored in this set of simulation - adaptive variants tend to be shared across more populations, even when their rate of substitution is higher (see the bottom-left panel - 100 times higher than the default rate of substitution explored in Figure 1 in the main text, and even in comparison to a simulation without substitutions, for the time frame explored in the upper-right panel).